

**Uptricks Services Pvt. Ltd.**

**AI Financial Fraud Detection &  
Preventive Action System**

*Internship Project – Report*

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**DOMAIN:** Artificial Intelligence

**PROJECT TYPE:** Machine Learning + FastAPI Web  
Application

## **Abstract**

Financial fraud is a major challenge for digital payment systems because suspicious transactions need to be identified quickly without unnecessarily affecting legitimate users. This project develops a machine-learning based transaction screening prototype that estimates fraud probability and converts the score into LOW, MEDIUM, or HIGH risk. These levels are mapped to ALLOW, REVIEW, and BLOCK actions. A FastAPI backend handles prediction and logging, while a web dashboard provides real-time visibility into transaction outcomes. The system is intended as an internship prototype and does not directly block real banking transactions.

## **1. Introduction**

The project was developed to understand how a machine-learning model can be turned into a usable financial-security application. Instead of stopping at model training, the work covers data, inference, decision logic, API development, user interface design, database logging, and monitoring.

## **2. Motivation**

A fraud model can produce a useful score, but a score alone is not enough for an operational system. Financial applications need a clear response to risk. This project therefore focuses on the complete chain from transaction screening to preventive action.

## **3. Objectives**

The objectives were to build a fraud prediction prototype, expose prediction through an API, provide understandable risk levels, demonstrate preventive actions, log transactions, and present model performance in a dashboard.

## **4. Methodology**

The methodology consists of model preparation, feature handling, prediction, threshold-based risk classification, preventive action mapping, database logging, and dashboard visualization. The Random Forest model produces a probability, and the decision layer converts that probability into an operational category.

## **5. System Architecture**

The architecture contains a browser frontend, FastAPI application layer, trained Random Forest model, preprocessing/feature transformation layer, SQLite database, and reporting files. The browser sends a transaction request to /predict. The backend validates the request, prepares the model input, predicts probability, decides risk and action, stores the event, and returns the result.

## **6. User-Friendly Input Design**

The original public fraud dataset contains anonymized V1–V28 features. Asking a normal user to enter these values would not be meaningful. The dashboard therefore uses fields such as amount, transaction type, location, device, and hour. The project separates user-friendly input from the internal model representation. This is an important usability improvement.

## **7. Results**

The dashboard demonstrates three decision categories. LOW corresponds to an allowed transaction, MEDIUM corresponds to a transaction held for review, and HIGH corresponds to a simulated block. Each result displays the probability and reason so that the reviewer can understand how the prediction is being used.

## **8. Limitations**

The system is a prototype. It does not have direct access to a bank or payment network and therefore cannot stop a real-world transaction. The public dataset uses anonymized features, which limits the ability to interpret individual V1–V28 components. Production fraud detection would also require additional behavioral, device, merchant, and velocity signals.

## **9. Future Scope**

Future work can include sandbox payment integration, stronger authentication, explainable AI, velocity checks, model drift monitoring, automated retraining, human review workflows, and production-grade security controls.

## **10. Conclusion**

The project successfully demonstrates an end-to-end machine-learning fraud-screening workflow. Its main strength is the connection between prediction and action: the model score is turned into a risk category and then into a preventive response. The project provides a practical foundation for extending fraud detection into a larger financial-security platform.

# 11. Output Screenshots

Figure 1 – Main dashboard

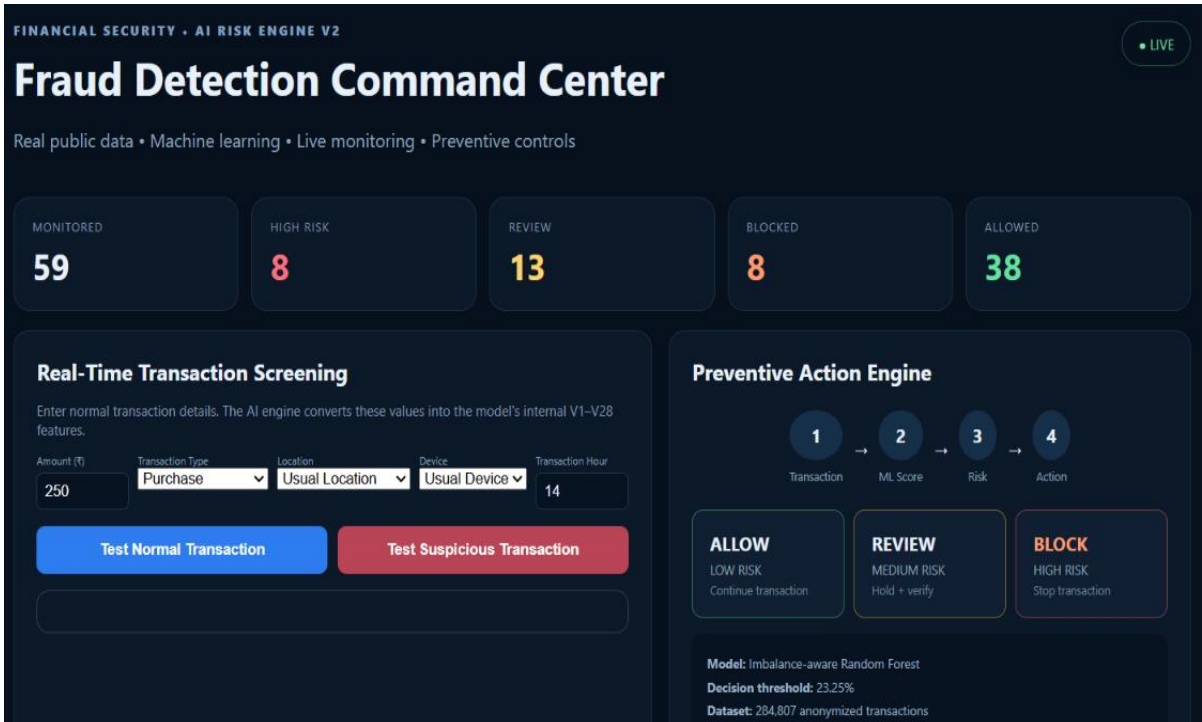


Figure 2 – LOW / ALLOW result

### Real-Time Transaction Screening

Enter normal transaction details. The AI engine converts these values into the model's internal V1-V28 features.

Amount (₹)  
250

Transaction Type  
Cash Withdrawal ▾

Location  
Usual Location ▾

Device  
New Device ▾

Transaction Hour  
14

Test Normal Transaction

Test Suspicious Transaction

| RISK LEVEL | FRAUD PROBABILITY | PREVENTIVE ACTION |
|------------|-------------------|-------------------|
| LOW        | 30.00%            | ALLOW             |

No strong fraud signal detected.

Transaction ALLOWED after automated risk screening.

Figure 3 – MEDIUM / REVIEW result

### Real-Time Transaction Screening

Enter normal transaction details. The AI engine converts these values into the model's internal V1–V28 features.

Amount (₹)  
1500

Transaction Type  
Purchase

Location  
Unusual Location

Device  
New Device

Transaction Hour  
2

Test Normal Transaction

Test Suspicious Transaction

RISK LEVEL

MEDIUM

FRAUD PROBABILITY

65.00%

PREVENTIVE ACTION

REVIEW

Suspicious transaction detected.

Transaction HELD for manual verification before settlement.

**Figure 4 – HIGH / BLOCK result**

MONITORED  
**64**

HIGH RISK  
**9**

REVIEW  
**15**

### Real-Time Transaction Screening

Enter normal transaction details. The AI engine converts these values into the model's internal V1-V28 features.

Amount (₹)  
60000

Transaction Type  
Money Transfer ▾

Location  
Unusual Location ▾

Device  
New Device ▾

Transaction Hour  
2

Test Normal Transaction

Test Suspicious Transaction

| RISK LEVEL  | FRAUD PROBABILITY | PREVENTIVE ACTION |
|-------------|-------------------|-------------------|
| <b>HIGH</b> | <b>105.00%</b>    | <b>BLOCK</b>      |

**Critical fraud probability detected.**  
Transaction BLOCKED to prevent potential financial loss.



Figure 5 – Transaction monitoring table

| Recent Transaction Monitoring |                     |            |                   |        |                   |           |
|-------------------------------|---------------------|------------|-------------------|--------|-------------------|-----------|
| ID                            | Time                | Amount     | Fraud probability | Risk   | Preventive action | Status    |
| TX-2AD22E226E                 | 2026-09-09 00:04:03 | ₹60000.00  | 105.00%           | HIGH   | BLOCK             | AUTOMATED |
| TX-7280A01F19                 | 2026-09-09 00:03:20 | ₹1500.00   | 65.00%            | MEDIUM | REVIEW            | AUTOMATED |
| TX-0991B16446                 | 2026-09-09 00:03:19 | ₹1500.00   | 65.00%            | MEDIUM | REVIEW            | AUTOMATED |
| TX-AES29C5324                 | 2026-09-09 00:02:00 | ₹250.00    | 30.00%            | LOW    | ALLOW             | AUTOMATED |
| TX-5ABEF54AF0                 | 2026-09-09 00:01:49 | ₹250.00    | 0.00%             | LOW    | ALLOW             | AUTOMATED |
| TX-95CAF526D2                 | 2026-09-08 23:05:12 | ₹15000.00  | 95.00%            | HIGH   | BLOCK             | AUTOMATED |
| TX-9005ECEC9C                 | 2026-09-08 23:05:08 | ₹15000.00  | 75.00%            | HIGH   | BLOCK             | AUTOMATED |
| TX-E91D91EE50                 | 2026-09-08 23:04:40 | ₹15000.00  | 38.28%            | MEDIUM | REVIEW            | AUTOMATED |
| TX-151A9F1215                 | 2026-09-08 23:03:34 | ₹250000.00 | 38.28%            | MEDIUM | REVIEW            | AUTOMATED |
| TX-FAF9988A91                 | 2026-09-08 23:03:33 | ₹250000.00 | 38.28%            | MEDIUM | REVIEW            | AUTOMATED |
| TX-45A90F4BA6                 | 2026-09-08 23:03:26 | ₹250000.00 | 38.28%            | MEDIUM | REVIEW            | AUTOMATED |
| TX-33FB1BAC4C                 | 2026-09-08 23:03:19 | ₹250000.00 | 30.52%            | LOW    | ALLOW             | AUTOMATED |
| TX-4E55A6C57D                 | 2026-09-08 23:03:18 | ₹250000.00 | 30.52%            | LOW    | ALLOW             | AUTOMATED |
| TX-AD30AF51A2                 | 2026-09-08 23:03:17 | ₹250000.00 | 30.52%            | LOW    | ALLOW             | AUTOMATED |
| TX-05DEDE1707                 | 2026-09-08 23:03:15 | ₹250000.00 | 30.52%            | LOW    | ALLOW             | AUTOMATED |
| TX-268A0A7286                 | 2026-09-08 23:03:07 | ₹250.00    | 4.98%             | LOW    | ALLOW             | AUTOMATED |

**Figure 6 – Model performance**

